Lab Assignment 9

Methods



CSE110: Programming Language I

No of Tasks			Points to Score
Classwork	**Evaluation**	**Homework**	**Homework** **4*10 = 40** **Assessment** **2*5 = 10**
2	**2**	**4**	

The students must complete the classwork tasks in the lab class to obtain the lab performance marks. They will also be marked based on the assessment tasks. The lab instructors may show/explain a few of the classwork tasks to the students if necessary. Any plagiarism in classwork or homework will lead to the student getting zero in the entire assignment. A random viva may take place.

Classwork

ClassWork 1

[A,B,C,D should be written in a single java file]

A. Write a method called **evenChecker** that takes an **integer** number as its argument and prints whether the number is even or odd **inside the method**.

Sample Method Call	Sample Output
`evenChecker(10);`	`Even!!`
`evenChecker(17);`	`Odd!!`

B. Write a method called **isEven** that takes an **integer** number as an argument and **returns** boolean true if the number is even otherwise **returns** boolean false.

Sample Method Call	Sample Output
`boolean result = isEven(10);` `System.out.println( result );`	`true`
`boolean result = isEven(17);` `System.out.println( result );`	`false`

C. Write a method called **isPos** that takes an **integer** number as an argument and **returns** boolean true if the number is positive otherwise **returns** boolean false.

Sample Method Call	Sample Output
`boolean result = isPos(-5);` `System.out.println( result );`	`false`
`boolean result = isPos(12);` `System.out.println( result );`	`true`

D. Write a method called **sequence()** that takes an **integer** in its parameter called n. Now, if n is **positive** then it prints all the **even** numbers from **0 to n**, otherwise if n is **negative** it prints all the **odd** numbers from **n to -1**.

Note: **You must call** the methods from **CW-1B** and **CW-1C**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
`sequence(10);`	`0 2 4 6 8 10`	Here, 10 is positive so 0,2,4,6,8,10 were printed.
`sequence(-7);`	`-7 -5 -3 -1`	Here, -7 is negative so -7,-5,-3,-1 were printed.
`sequence(7);`	`0 2 4 6`	Here, 7 is positive so 0,2,4,6 were printed
`sequence(-8);`	`-7 -5 -3 -1`	Here, -8 is negative so -7,-5,-3,-1 were printed.

ClassWork 2

[A,B,C should be written in a single java file]

A. Write a method called **circleArea** that takes an **integer** radius in its parameter and **returns** the **area** of the circle.

Note: area of a circle is πr^2

Sample Method Call	Sample Output
`double area = circleArea(5);` `System.out.println(area);`	`78.5398`

B. Write a method called **sphereVolume** that takes an **integer** radius in its parameter and **returns** the **volume** of the sphere.

Note: volume of a sphere is $\frac{4}{3}\pi r^3$

Sample Method Call	Sample Output
`double volume = sphereVolume(5);` `System.out.println(volume);`	`523.5987`

C. Write a method called **findSpace** that takes two values in its parameters one is an **integer** diameter and another one is a String. Using the given diameter, this method should calculate the Area of a circle or the Volume of a sphere depending on the value of the second parameter. Finally, it should print the result **inside the method**.

Note: **You must call** the method written in task **CW-2A & CW-2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output
`findSpace(10,"circle");`	`78.5398`
`findSpace(10,"sphere");`	`523.5987`
`findSpace(10,"square");`	`“Wrong Parameter”`

Evaluation

[A,B should be written in a single java file]

A. Write a method called **isTriangle** that takes 3 integer numbers as arguments. The method will **return** the boolean True if the 3 sides can form a valid triangle otherwise it'll **return** the boolean False.

Note: In a valid triangle, the sum of **any** two sides will be greater than the third side.

Sample Method Call	Sample Output	Explanation
`boolean res = isTriangle(7,5,10);` `System.out.println( res );`	`true`	Here, 7+5>10, 5+10>7 also, 10+7>5. Thus, these 3 sides can form a valid triangle.
`boolean res = isTtriangle(3,2,1);` `System.out.println( res );`	`false`	Here, 1+2<=3, thus, these 3 sides can NOT form a valid triangle.

B. Write a method called **triArea** that takes 3 sides of a triangle as 3 **integer** arguments. The method should calculate and print the area of the triangle only if it's a valid triangle otherwise print that it's not a valid triangle.
Area of triangle = √[s(s−a)(s−b)(s−c)], where 's' is the semi perimeter of the triangle. So, semi-perimeter = s = perimeter/2 = (a + b + c)/2.

Note: **You must call** the method written in task **Evaluation-A**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output	Explanation
`triArea(3,2,1);`	`Can't form triangle`	Here, 1+2<=3, thus, these 3 sides can NOT form a valid triangle.
`triArea(7,5,10);`	`16.248`	Here, 7,5,10 is able to form a valid triangle so, using the formula we get the area as 16.248

HomeWork

HomeWork 1

A. Write a method called **isPrime** which takes an integer in its parameter to check whether a number is prime or not. If the number is prime then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
`boolean check = isPrime(7);` `System.out.println(check);`	`true`
`boolean check = isPrime(15);` `System.out.println(check);`	`false`

B. Write a method called **isPerfect** which takes an integer in its parameter to check whether a number is perfect or not. If the number is perfect then the method returns boolean **true** otherwise it returns boolean **false**.

Sample Input	Sample Output
`boolean check = isPerfect(6);` `System.out.println(check);`	`true`
`boolean check = isPerfect(33);` `System.out.println(check);`	`false`

C. Write a method called **special_sum** that calculates the sum of all numbers that are either prime numbers or perfect up till the integer value given in its parameter. This integer value must be taken as user input and passed into the method.

Note: **You must call** the methods written in task **HW-1A & HW-1B**, otherwise this task will be **considered invalid**.

Sample Input	Sample Output	Output
8	`int result = special_sum(8);` `System.out.println(result);`	23
Explanation:	Between 1 to 8 the Prime numbers are 2,3,5,7 and 6 is a Perfect number. So, the summation is 2+3+5+7+6=23.	

HomeWork 2

A. Write a simple method called **showDots** that takes a number as an argument and then prints that amount of dots inside the method.

Note: You can use `System.out.print()` to avoid the next output being printed on the next line.

Sample Method Call	Sample Output
`showDots(5);`	.....
`showDots(3);`	...

B. Write a method called **show_palindrome** that takes a number as an argument and then prints a palindrome inside the method.

Note: You can use `System.out.print()` to avoid the next output being printed on the next line

Sample Method Call	Sample Output
`show_palindrome(5)`	123454321
`show_palindrome(3)`	12321

C. Write a method called **showDiamond** that takes an integer number as an argument and then prints a **palindromic diamond shape**. Moreover, the empty spaces surrounding the diamonds are filled with dots(**.**) .
Note: **You must call** the methods written in task **HW-2A & HW-2B**, otherwise this task would be **considered invalid**.

Sample Method Call	Sample Output

`showDiamond(5)`	`....1....`
	`...121...`
	`..12321..`
	`.1234321.`
	`123454321`
	`.1234321.`
	`..12321..`
	`...121...`
	`....1....`

`showDiamond(3)`	`..1..`
	`.121.`
	`12321`
	`.121.`
	`..1..`

HomeWork 3

A. Write a method called **calcTax** that takes 2 arguments which are **your age** then **your salary**. The method must calculate and **return** the tax as per the following conditions:

- No tax if you are less than 18 years old.
- No tax if you get paid less than 10,000
- 7% tax if you get paid between 10K and 20K
- 14% tax if you get paid more than 20K

Sample Method Call	Output	Explanation
`double t = calcTax(16,20000);` `System.out.println(t);`	`0.0`	Here, the age is less than 18 so 0 tax.
`double t = calcTax(20,18000);` `System.out.println(t);`	`1260.0`	Here, the age is greater than 18 and income is between 10K-20K so tax is 7% of 18000 = 1260.

B. Write a method called **calcYearlyTax** that takes no arguments. Inside the method it should take **first input as your age** and then **12 other inputs** as income of each month of the year. The method must calculate and print Tax for each month and finally print the total Tax of the whole year based on the **HW-3A conditions**.

Note: You must call the method written in task **HW-3A**, otherwise this task would be **considered invalid**.

Sample Method Call	Output	Explanation
calcYearlyTax()	22 8000 15000 22000 2300 15300 21000 34000 9000 27000 88000 32000 7300	Month1 tax: 0 Month2 tax: 1050.0 Month3 tax: 3080.0 Month4 tax: 0 Month5 tax: 1071.0 Month6 tax: 2940.0 Month7 tax: 4760.0 Month8 tax: 0 Month9 tax: 3780.0 Month10 tax: 12320.0 Month11 tax: 4480.0 Month12 tax: 0 Total Yearly Tax: 33481.0

HomeWork 4

Write a method called **oneToN** that prints 1 till N recursively.

Hint: N is a number taken as input from the user and you need to print the numbers starting from 1 to N recursively.

Sample Input	Sample Method Call	Output
N=5	oneToN(1,N);	1 2 3 4 5
N=11	oneToN(1,N);	1 2 3 4 5 6 7 8 9 10 11